Unit – II
Wave Mechanics & X-Ray Diffraction

(1)

When classical theory fails to give a satisfactory explanation about the phenomenon of photoelectric effect and Compton Effect etc. Max Planck in 1901 gives a new concept called plank's hypothesis. According to this:

- A black body consists of a large number of oscillators vibrating with all possible frequencies.
- The oscillator of frequency v can exist in states whose energy E is an integral multiple of a constant h and the frequency v of the oscillator is

$$E = nh\nu;\ n = 1, 2, 3\ldots$$

- The oscillator emits energy only when it passes from higher energy level to lower energy level.

Wave-Particle duality of Matter

Light when considered as a wave based on interference, diffraction and polarization prove the wave nature of radiations because they require two waves at the same position at the same time. On the other hand, it is impossible for two particles to occupy the same position at the same time. But for the experiments like photoelectric effect or Compton Effect radiation is considered as particle and interacts with matter. So, there are two contradictory aspects:

- A wave spreads out and occupies a relatively large space.
- A particle occupies a definite position in space and hence occupies small space.

If one has to explain the results of the experiments performed with radiation then it becomes necessary to accept this contradictory nature of light.

De-Broglie Matter Waves

In 1924, De-Broglie said that matter has dual characteristic just like radiation. His concept was based on his observations:

- The whole universe is composed of matter and electromagnetic radiations. Since both are forms of energy so can be transformed into each other.
- The nature loves symmetry. As the radiation has dual nature, matter should also posses dual character.

The waves associated with moving particles are matter waves or de-Broglie waves.

Wavelength of de-Broglie Matter waves

Consider a photon whose energy is given by

$$E = h\nu = hc/\lambda \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

If a photon posses mass, then according to the theory of relativity, its energy is given by

$$E = mc^2 \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

From (1) & (2)

$$\text{Mass of photon } m = h/c\lambda$$

Therefore, Momentum of photon

$$p = mc = (h/c\lambda).c = h/\lambda$$

If instead of a photon, we consider a material particle of mass m moving with velocity v, then the momentum of the particle, p = mv. Therefore, the wavelength of the wave associated with the moving particle is given by,

$$\lambda = h/mv$$

This is called de-Broglie wavelength.

Different Expressions:

a) If E is the kinetic energy of a material, then
$E = mv^2/2 = P^2/2m$

Or $$P = \sqrt{2mE}$$

Therefore, the De-Broglie wavelength λ...

$$\lambda = \frac{h}{\sqrt{2mE}}$$

b) When a charge particle carrying a charge q is accelerated by potential difference V volts, then its kinetic energy is

$$E = qV$$

Therefore, the De-Broglie wavelength λ...

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

c) When a material particle is in thermal equilibrium at a temperature T, then

$$E = \frac{3}{2}kT$$

Where k is Boltzmann's constant = $1.38\times10^{-23} J/K$
Therefore, the De-Broglie wavelength λ...

$$\lambda = \frac{h}{\sqrt{2m(\frac{3}{2}kT)}} = \frac{h}{\sqrt{3mkT}}$$

d) If the velocity of the particle is comparable with the velocity of light, then the mass of the particle is given by

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

Where m_0 is the rest mass of the particle.
Therefore, the De-Broglie wavelength λ...

$$\lambda = \frac{h}{mv} = \frac{h\sqrt{1-\frac{v^2}{c^2}}}{m_0 v}$$

e) **De-Broglie wavelength associated with electrons:**
Let us consider the case of an electron of rest mass m_0 and charge e which is accelerated by a potential V volt from rest to velocity v, then

$$\frac{1}{2}mv^2 = eV \text{ or } v = \sqrt{\frac{2eV}{m_0}}$$

If the relativistic variation of mass with velocity of the electron is ignored, then

$$m \approx m_0 \text{ and } \lambda = \frac{h}{m_0 v}$$

So,
$$\lambda = \frac{h\sqrt{m_0}}{m_0\sqrt{2eV}} = \frac{h}{\sqrt{2eVm_0}}$$

Or,

$$\lambda = \frac{6.625\times10^{-34}}{\sqrt{2\times1.632\times10^{-10}V\times9.1\times10^{-31}}} \Rightarrow \lambda = \frac{12.26}{\sqrt{V}}A^0$$

If V=100 volt, then

$$\lambda = 1.226A^0$$

Therefore, the wavelength associated with an electron accelerated to 100 volt is 1.226 Å.

Properties of Matter Waves

- Lighter is the particle, greater is the wavelength associated with it.
- Smaller is the velocity of particle, greater is the wavelength associated with it.
- When $v = 0$ then $\lambda = \infty$, i. e., wave becomes indeterminate and if $v = \infty$ then $\lambda = 0$. This shows that matter waves are generated by the motion of the particles. These waves are produced whether the particle are charged or uncharged. This implies that these waves are not electromagnetic but they are a new kind of waves.
- The velocity of matter waves depends upon the velocity of matter particle.
- The velocity of matter waves is greater than the velocity of light................

  The energy of a wave of frequency ν is given by $E = h\nu$

  The energy of a particle of mass m is given by $E = mc^2$

$$h\nu = mc^2 \text{ or } \nu = \frac{mc^2}{h}$$

  The wave velocity ω is $\omega = \nu\lambda$

  So, $\omega = \frac{\nu h}{mv}$, where v is the velocity of particle.

  So, $\boxed{\omega = \frac{c^2}{v}}$

- As particle velocity v cannot exceed c, ω is greater than velocity of light. Therefore, the velocity of matter waves always greater than c. This shows that matter waves are not physical waves.

Wave Velocity & Group Velocity

Whenever we talk about a particle, we think of a point object which occupies a definite position in space and which has definite momentum but in quantum mechanics particle can be described by a wave packet.

A wave packet comprises a group of waves slightly differing in velocity and wavelength, with phases and amplitude such that they interfere constructively over a small region of space where the particle can be located and outside this space they interfere destructively so that the amplitude reduces to zero.

The velocity of component waves of a wave packet is called as *phase velocity* of those waves.
The velocity, with which the wave packet travels, is called *group velocity.*

Expression for group velocity

Consider two wave trains having same amplitude a but slightly different angular frequencies (ω and ω') and phase velocities (u and u'). The waves can be represented as

$$y_1 = a\sin(\omega t - kx) \quad ....(1)$$

$$y_2 = a\sin(\omega' t - k'x) \quad ....(2)$$

Where k and k′ are propagation constants defined as (2π/λ). The resultant wave is

$$y = y_1 + y_2 = a\sin(\omega t - kx) + a\sin(\omega' t - k'x)$$

$$= 2a\cos\left[\left(\frac{\omega-\omega'}{2}\right)t - \left(\frac{k-k'}{2}\right)x\right] \times \sin\left[\left(\frac{\omega+\omega'}{2}\right)t - \left(\frac{k+k'}{2}\right)x\right]$$

$$\approx 2a\cos\left[\left(\frac{d\omega}{2}\right)t - \left(\frac{dk}{2}\right)x\right]\sin(\omega t - kx) \ldots\ldots \left(\because \sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)\right) \ldots\ldots\ldots\ldots (3)$$

Where $\frac{\omega+\omega'}{2} = \omega$ and $\frac{k+k'}{2} = k(approx.)$

Eq. (3) represents a wave of angular frequency ω and propagation constant k. The phase velocity v_p o the resultant wave is given by

$$\boxed{v_p = \frac{\omega}{k}}$$

The amplitude of the resultant wave is modified.

$$2a\cos\left[\left(\frac{d\omega}{2}\right)t - \left(\frac{dk}{2}\right)x\right] \text{ or } 2a\cos\frac{d\omega}{2}\left[t - \frac{dk}{d\omega}x\right] = 2a\cos\frac{d\omega}{2}\left[t - \frac{x}{v_g}\right]$$

Where v_g is called group velocity.

$$\boxed{v_g = \frac{d\omega}{dk} = \frac{\omega-\omega'}{k-k'}}$$

Relation between group velocity and phase velocity in dispersive medium

$$\omega = v_p k \text{ or } d\omega = dv_p k + v_p dk$$

$$\frac{d\omega}{dk} = v_p + k\frac{dv_p}{dk} \text{ or } v_g = v_p + k\frac{dv_p}{dk}$$

Since, $k = 2\pi/\lambda$ hence,

$$v_g = v_p + \frac{2\pi}{\lambda}\frac{dv_p}{d\left(\frac{2\pi}{\lambda}\right)} = v_p + \frac{1}{\lambda}\frac{dv_p}{d\left(\frac{1}{\lambda}\right)}$$

$$\because\ d\left(\frac{1}{\lambda}\right) = -\frac{1}{\lambda^2}d\lambda$$

$$\boxed{v_g = v_p - \lambda\frac{dv_p}{d\lambda}}$$

a dispersive medium wave velocity or phase velocity is frequency dependent.

In Non-dispersive medium or free space, $\frac{dv_p}{d\lambda} = 0$

Therefore, $\boxed{v_g = v_p}$

Group Velocity & Particle Velocity

$$v_g = v_p - \lambda\frac{dv_p}{d\lambda}$$

It can be written as

$$v_g = \lambda^2\left[\frac{v_p}{\lambda^2} - \frac{1}{\lambda}\frac{dv_p}{d\lambda}\right]$$

$$= -\lambda^2\frac{d}{d\lambda}\left(\frac{v_p}{\lambda}\right) = -\lambda^2\frac{dv}{d\lambda}$$

Or

$$\frac{1}{v_g}=-\frac{1}{\lambda^2}\frac{d\lambda}{d\nu}=\frac{d}{d\nu}\left(\frac{1}{\lambda}\right)$$

If E and V represent the total and potential energy of the particle respectively, then kinetic energy of particle is given by

$$\frac{1}{2}mv^2=E-V \quad \text{where v is the particle velocity}$$

Or
$$v=\left[\frac{2(E-V)}{m}\right]^{1/2}$$

According to de-Broglie formula $\lambda = h/mv$

$$\therefore \frac{1}{\lambda}=\frac{mv}{h}=\frac{m}{h}\left[\frac{2(E-V)}{m}\right]^{1/2}$$

$$\frac{1}{v_g}=\frac{d}{d\nu}\left(\frac{m}{h}\left[\frac{2(E-V)}{m}\right]^{1/2}\right)$$

$$=\frac{d}{d\nu}\left(\frac{m}{h}\left[\frac{2(h\nu-V)}{m}\right]^{1/2}\right)\ldots\ldots\ldots\ldots\ldots\ldots\ldots(E=h\nu)$$

$$=\frac{1}{h}\frac{d}{d\nu}\left[(2m(h\nu-V))^{1/2}\right]$$

$$=\frac{1}{h}\frac{1}{2}(2m(h\nu-V))^{1/2}.2mh$$

$$=\frac{m}{(2m(h\nu-V))^{1/2}}=\left[\frac{m}{2m(E-V)}\right]^{1/2}=\frac{1}{v_p}$$

Therefore, $\boxed{\text{Group velocity } v_g = \text{particle velocity } v_p}$

Hence, we can say that a material particle is in motion is equivalent to group of waves or a wave packet.

Phase velocity of de-Broglie waves

According to de-Broglie, a particle of mass **m** moving with velocity **v** is associated with a wave whose wavelength λ is given by

$$\lambda=\frac{h}{mv}$$

The propagation constant k of the wave is given by

$$k=\frac{2\pi}{\lambda}=\frac{2\pi(mv)}{h}$$

Let E be the energy of the particle, $E=h\nu \Rightarrow \nu=\frac{E}{h}$

Therefore, angular frequency...... $\omega = 2\pi\nu = 2\pi E/h$

$$\omega=\frac{2\pi mc^2}{h}\ldots\ldots\ldots\ldots(\because E=mc^2)$$

De-Broglie phase or wave velocity v_p is given by

$$v_p=\frac{\omega}{k}=\left(\frac{2\pi mc^2}{h}\right)\times\frac{h}{2\pi(mv)}=\frac{c^2}{v}$$

As **v** is less than **c**, de-Broglie wave velocity must be greater than **c**. so, the de-Broglie wave train associated with the particle would travel much faster than the particle itself and would leave the particle far behind. This statement is nothing but collapse of the wave description of the particle.

Group velocity of de-Broglie waves

The angular frequency and propagation constant of de-Broglie wave associated with a particle of rest mass $\mathbf{m_0}$ can be calculated as.....

According to theory of relativity,

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

We know that $\omega = \frac{2\pi mc^2}{h}$ and $k = \frac{2\pi}{\lambda} = \frac{2\pi(mv)}{h}$

So, on substituting the value of **m**

$$\omega = \frac{2\pi m_0 c^2}{h\sqrt{1-\frac{v^2}{c^2}}} \text{ and } k = \frac{2\pi m_0 v}{h\sqrt{1-\frac{v^2}{c^2}}}$$

Differentiating these eqns. We have

$$\frac{d\omega}{dv} = \frac{2\pi m_0 v}{h\left(1-\frac{v^2}{c^2}\right)^{3/2}} \text{ and } \frac{dk}{dv} = \frac{2\pi m_0}{h\left(1-\frac{v^2}{c^2}\right)^{3/2}}$$

The group velocity v_g of de-Broglie waves associated with the particle is given by

$$v_g = \frac{d\omega}{dk} = \frac{d\omega / dv}{dk / dv}$$

Substituting the values of $(d\omega / dv)$ and (dk / dv), we get

$$\boxed{v_g = v}$$

Thus, **de-Broglie wave** group associated with moving particle **travels with same velocity** as **the particle.**

Since, $v_p = \frac{c^2}{v}$

Therefore, wave velocity and group velocity of de-Broglie wave is related as

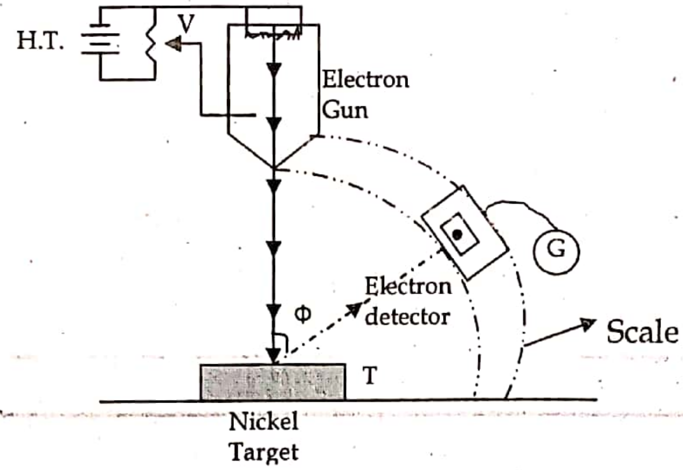
$$v_p . v = c^2$$

Or

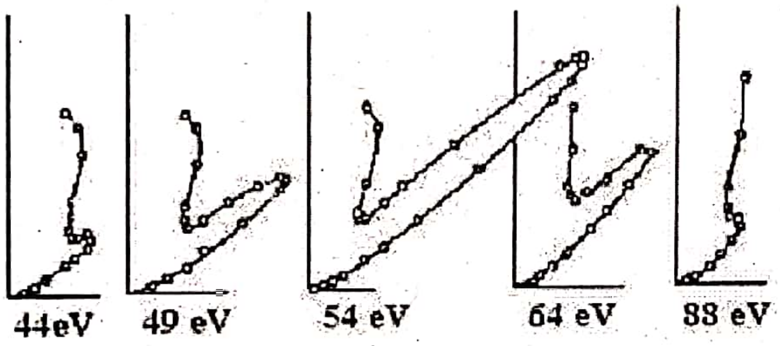
$$\boxed{v_p . v_g = c^2}$$

Davisson & Germer's electron diffraction experiment

De-Broglie said that material particles have wave like character, & they are expected to show the phenomenon like interference and diffraction. In 1927, Davisson and Germer prove the existence of de-Broglie waves. In his experiment electrons are produced by thermionic emission from a tungsten filament mounted in an electron gun. The ejected electrons are accelerated towards anode in an electric field of known potential difference and collimated into a narrow beam. The narrow beam of electrons is allowed to fall on the surface of a nickel crystal. Nickel is used as target because its atoms are arranged in regular manner so the surface of crystal acts as a diffraction grating. The electrons scattered from the target are collected by a Faraday cylinder called electron detector which is connected to a sensitive galvanometer and can be moved along a circular scale.



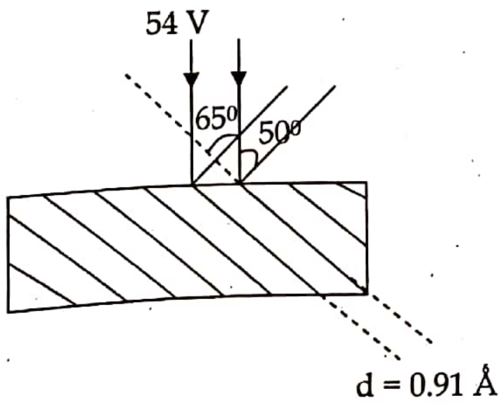
For different values of potential the electrons were collected at different positions. The current which is the measure of the intensity of the diffracted beam, is plotted against the diffraction angle Φ for each accelerating potential.



It is observed that at the voltage of 40 volts, a smooth curve is obtained and a bump begins to appear in the curve for 44 volts electrons. As the potential difference is further increased the bump start shifting upward and becomes most prominent for 54 volts electrons at $\Phi = 50^0$. Beyond 54 volts the bump gradually diminishes and become insignificant at 68 volts electrons. The peak at 54 volts and 50^0 provided the evidence that electron were diffracted and verifies the existence of electron wave.
According to de-Broglie, the wavelength associated with electron accelerated through a potential V is given by

$$\frac{12.26}{\sqrt{V}}\overset{0}{A} = \frac{12.26}{\sqrt{54}}\overset{0}{A} = 1.67\overset{0}{A}$$

From X-ray analysis, it is known that nickel crystal acts as plane diffraction grating with grating space d = 0.91 Å. According to experiment we have diffracted electron beam at $\Phi = 50^0$. The corresponding angle of incidence relative to the family of Bragg plane



$$\theta = \frac{180-50}{2} = 65^0$$

Using Bragg's equation (taking n = 1). We get

$$\lambda = 2d\sin\theta = 2(0.91)\sin 65^0 = 1.65\overset{0}{A}.$$

This is in good agreement with the wavelength computed from de-Broglie hypothesis. Hence, confirms the de-Broglie concept of matter waves.

Heisenberg's Uncertainty Principle

In 1927, Heisenberg proposed a very interesting principle, which is a direct consequence of the dua nature of matter, known as uncertainty principle or to determine the exact position and momentum simultaneously Heisenberg proposed the principle of uncertainty.

"It is impossible to measure precisely and simultaneously both the members of pairs of certain canonically conjugate variables that describe the behaviour of an atomic system."

Canonically conjugate pairs of physical quantities are those in which measurement of one quantity affects the capacity to measure the other.

"The product of uncertainties in determining the position and momentum of a particle at the same instant is of the order of $\hbar$."

Uncertainty in momentum Δp_x, and in position Δx

$$\Delta p_x . \Delta x \geq \hbar \approx \frac{h}{2\pi}$$

Uncertainty in angular momentum and angular position

$$\Delta J . \Delta \phi \geq \hbar \approx \frac{h}{2\pi}$$

Uncertainty in kinetic energy and time

$$\Delta T . \Delta t \geq \hbar \approx \frac{h}{2\pi}$$

The exact principle is stated as *"The product of uncertainties in determining the position and momentum of the particle can never be smaller than the number of the order $\frac{\hbar}{2}$."*

Physical Significance of Heisenberg's Uncertainty Principle

- This principle explains why it is possible for radiation and matter to have dual nature.
- It helps in understanding many phenomenon like absence of electrons within nucleus, existence of protons & neutrons in nucleus, binding energy of an electron in atom etc.
- It also states that we can only predict the probable behaviour of quantum mechanical systems & not the exact behaviour.

Applications of Heisenberg Uncertainty Principle

- Non-Existence of electrons in the nucleus
- Radius of Bohr's First orbit
- Binding energy of an electron in an atom
- Zero point energy of harmonic oscillator

(You have to do these applications yourself.)

Nave Function

\ccording to uncertainty principle, the position and momentum of a particle cannot be measured ccurately at the same time; the measurement of one quantity introduces an uncertainty into the other. n case of electromagnetic waves, the electric and magnetic fields varies periodically. In water waves, eight of the water surface varies periodically. Similarly in sound waves, pressure varies periodically. o, what varies in matter waves? Answer is ....*Wave function (Ψ).*

:hrodinger describes the amplitude of matter waves by a complex quantity $\Psi(x, y, z, t)$ known as wave ınction.

Physical significance of wave function

- The value of wave function associated with a moving particle at a particular point (x, y, z) in space at the time 't' is related to the possibility of finding the particle there at that time.
- The physical significance of wave function is that the square of its absolute value $|\psi|^2$ at a point is proportional to the probability of finding the particle described by the wave function in a small element of volume $d\tau$ ($dxdydz$) at that point.
- $|\psi|^2$ gives the probability of finding the particle at that point at any given moment. $|\psi|^2$ is called probability density and ψ is called probability amplitude. Particle is necessarily somewhere in space, therefore

$$\int_{-\infty}^{\infty} |\psi|^2 \, d\tau = 1$$

- A wave function ψ satisfying this relation called normalized wave function.

Nature of wave function

- It must be finite everywhere.
- It must be single valued.
- It must be continuous.

Schrodinger time independent wave equation (Method 1)

According to de-Broglie theory, a particle of mass m always associated with a wave whose wavelength is $\lambda = h/mv$. If particle is a wave it is expected that there should be some sort of wave equation which describes the behaviour of the particle. Consider a system of stationary waves associated with a particle. Let x, y, z be the co-ordinates of particle and ψ is the wave displacement at any time t. ψ is finite, single valued and periodic function. The classical differential equation of wave motion is expressed as......

$$\frac{\partial^2\psi}{dt^2} = v^2\left(\frac{\partial^2\psi}{dx^2} + \frac{\partial^2\psi}{dy^2} + \frac{\partial^2\psi}{dz^2}\right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

$$\frac{\partial^2\psi}{dt^2} = v^2\nabla^2\psi \ldots\ldots\ldots\ldots \left[\because \nabla^2 = \left(\frac{\partial^2\psi}{dx^2} + \frac{\partial^2\psi}{dy^2} + \frac{\partial^2\psi}{dz^2}\right)\right] \ldots\ldots\ldots\ldots(\because \nabla^2 = laplacian\ operator)$$

Solution of eqn (1) is given by...

$$\psi = \psi_0 \sin \omega t = \psi_0 \sin 2\pi v t \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

$$\frac{d\psi}{dt} = \psi_0(2\pi v t)\cos 2\pi v t \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(3)$$

$$\frac{d^2\psi}{dt^2} = -\psi_0(2\pi v t)^2 \sin 2\pi v t = -4\pi^2 v^2 \psi = -\frac{4\pi^2 v^2}{\lambda^2}\psi \ldots\ldots(4)$$

Putting value of $\frac{d^2\psi}{dt^2}$ in equation (1)

$$v^2\nabla^2\psi = -\frac{4\pi^2 v^2}{\lambda^2}\psi$$

$$\nabla^2\psi + \frac{4\pi^2}{\lambda^2}\psi = 0$$

$$\nabla^2\psi + \frac{4\pi^2}{h^2}m^2v^2\psi = 0 \ldots\ldots\ldots(\because \lambda = h/mv)$$

If E and V be the total and potential energies of particle, then its kinetic energy

$$\frac{1}{2}mv^2 = E - V$$

In this case it should be remembered that potential energy is independent of time.

$$m^2v^2 = 2m(E-V)$$

$$\text{Therefore, } \nabla^2\psi + \frac{8\pi^2}{h^2}m(E-V)\psi = 0$$

$$\boxed{\nabla^2\psi + \frac{2m}{\hbar^2}(E-V)\psi = 0} \text{..........}(\because h = 2\pi\hbar)$$

This is *Schrodinger time independent wave equation.*
For a free particle, V=0

$$\boxed{\nabla^2\psi + \frac{2m}{\hbar^2}E\psi = 0}$$

Method 2

The differential equation for a free particle in one dimension is given by

$$\frac{\partial^2\psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} \text{-----------------------------------(1)}$$

Standard solution of this equation is

$$\psi = \psi_0 e^{-i\omega t}$$

$$\frac{d\psi}{dt} = -i\omega\psi_0 e^{-i\omega t}$$

$$\frac{d^2\psi}{dt^2} = (-i\omega)(-i\omega)\psi_0 e^{-i\omega t} = -\omega^2\psi_0 e^{-i\omega t}$$

Putting the value of $\frac{d^2\psi}{dt^2}$ in eqn (1). We get,

$$\frac{d^2\psi}{dx^2} = -\frac{\omega^2}{v^2}\psi_0 e^{-i\omega t}$$

$$\frac{d^2\psi_0}{dx^2}e^{-i\omega t} = -\frac{\omega^2}{v^2}\psi_0 e^{-i\omega t} \Rightarrow \frac{d^2\psi_0}{dx^2} = -\frac{\omega^2}{v^2}\psi_0$$

$$\frac{d^2\psi_0}{dx^2} = -\frac{(2\pi\nu)^2}{v^2}\psi_0 = -\frac{4\pi^2}{\lambda^2}\psi_0 \text{...........}(\because \omega = 2\pi\nu, \nu = v/\lambda)$$

$$\frac{d^2\psi_0}{dx^2} = -\frac{4\pi^2 m^2 v^2}{h^2}\psi_0 \text{...........}(\because \lambda = h/mv)$$

If E and V be the total and potential energies of particle, then its kinetic energy

$$\frac{1}{2}mv^2 = E - V$$

In this case it should be remembered that potential energy is independent of time.

$$m^2v^2 = 2m(E-V)$$

So,

$$\frac{d^2\psi_0}{dx^2} = -\frac{8\pi^2 m^2}{h^2}(E-V)\psi_0 = -\frac{2m}{\hbar^2}(E-V)\psi_0$$

Or

$$\frac{d^2\psi_0}{dx^2} + \frac{2m}{\hbar^2}(E-V)\psi_0 = 0$$

In 3-D,

$$\boxed{\nabla^2\psi_0 + \frac{2m}{\hbar^2}(E-V)\psi_0 = 0}$$

Schrodinger time dependent wave equation

The differential equation for a free particle in one dimension is given by

$$\frac{\partial^2\psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

Standard solution of this equation is

$$\psi(r,t) = Ae^{-i2\pi(\nu t - \nu\frac{x}{v})}$$

$$= Ae^{-i2\pi(\frac{E}{h}t - \frac{x}{\lambda})} \text{.....................}(\because \nu = \frac{E}{h}, v = \nu\lambda)$$

$$= Ae^{-i2\pi(\frac{E}{h}t - \frac{px}{h})} \text{.....................}(\because \lambda = \frac{h}{p})$$

$$\psi(r,t) = Ae^{-i\frac{2\pi}{h}(Et-px)} = Ae^{-\frac{i}{\hbar}(Et-px)} \text{............}(1)$$

This represents a particle moving along the x direction having total energy E and momentum p.
Differentiating (1) wrt t.

$$\frac{\partial\psi}{\partial t} = -\frac{i}{\hbar}EAe^{-i\frac{2\pi}{h}(Et-px)} = -\frac{i}{\hbar}E\psi$$

$$E\psi = -\frac{\hbar}{i}\frac{\partial\psi}{\partial t} = i\hbar\frac{\partial\psi}{\partial t} \text{...................}(2)$$

Differentiating (1) wrt x twice

$$\frac{\partial^2\psi}{\partial x^2} = \left(-\frac{i}{\hbar}\right)^2(-p)^2Ae^{-i\frac{2\pi}{h}(Et-px)} = -\frac{p^2}{\hbar^2}Ae^{-i\frac{2\pi}{h}(Et-px)} = -\frac{p^2}{\hbar^2}\psi$$

$$p^2\psi = -\hbar^2\frac{\partial^2\psi}{\partial x^2} \text{.........................}(3)$$

Let E and U be the total and potential energies of the particle.

So,
$$E = K + U$$
$$= \frac{p^2}{2m} + U$$

Multiply ψ on both sides

$$E\psi = \frac{p^2\psi}{2m} + U\psi$$

Substituting the values of $E\psi$ and $p^2\psi$. We get,

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + U\psi}$$

This is time dependent Schrodinger equation in one dimension.

In 3D, time dependent Schrodinger equation is

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + U\psi \text{ ....}(\because \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$$

Schrodinger time independent equation from time dependent equation

The differential equation for a free particle in one dimension is given by

$$\frac{\partial^2\psi}{\partial x^2}=\frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

Standard solution of this equation is

$$\psi = Ae^{-\frac{i}{\hbar}(Et-px)} = Ae^{-\frac{i}{\hbar}Et}e^{\frac{ipx}{\hbar}} = \psi_0 e^{-\frac{i}{\hbar}Et}\;\ldots\ldots\ldots\ldots\ldots\ldots(\because Ae^{\frac{ipx}{\hbar}}=\psi_0)$$

Differentiating ψ wrt t.We get,

$$\frac{d\psi}{dt}=-\frac{i}{\hbar}E\psi_0 e^{-\frac{i}{\hbar}Et}$$

Differentiating ψ wrt x twice. We get,

$$\frac{d^2\psi}{dx^2}=\frac{d^2\psi_0}{dx^2}e^{-\frac{i}{\hbar}Et}$$

The time dependent Schrodinger eqn is,

$$i\hbar\frac{\partial\psi}{\partial t}=-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}+U\psi$$

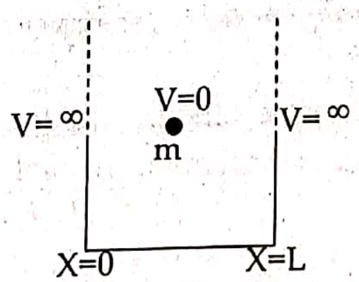
Putting the values of ψ, $\frac{d\psi}{dt}$ and $\frac{d^2\psi}{dx^2}$. We get

$$\boxed{\frac{d^2\psi_0}{dx^2}+\frac{2m}{\hbar^2}(E-U)\psi_0=0}$$

The values of energy for which the time independent Schrodinger equation can be solved are called Eigen values and the corresponding function is called Eigen function.

Particle in one dimension box with infinitely hard walls

Consider a particle of mass **m** moving along x-axis between the two rigid wall A & B at x=0 & x=L. Particle is free to move between the walls. The potential energy of the particle between the two walls is constant because no force is acting on the particle. The constant potential energy is taken to be zero for simplicity. The particle does not loose energy when it strikes back and forth in the potential well because the walls are infinitely rigid.



V(x) =∞ for x≤0 and x≥L

V(x) =0 for 0<x<L

The time independent Schrodinger equation for the particle

$$\frac{d^2\psi}{dx^2}+\frac{2m}{\hbar^2}(E-V)\psi=0$$

$$\frac{d^2\psi}{dx^2}+\frac{2mE}{\hbar^2}\psi=0\ldots\ldots\ldots(\because V=0)$$

$$\frac{d^2\psi}{dx^2}+K^2\psi=0\ldots\ldots\ldots(\because K^2=\frac{2mE}{\hbar^2})$$

It is a form of standard differential equation.

Solution of the equation is given by

$$\psi(x) = A\sin Kx + B\cos Kx$$

Where A and B are the constants. Values of these constants can be obtained by applying boundary conditions.

$$\psi = 0,\ \text{At } x=0\ (\because \text{ particle cannot penetrate the walls})$$

$$0 = A\sin 0 + B\cos 0 \Rightarrow B\cos 0 = 0 \Rightarrow B = 0$$

Therefore the term **B cos Kx** can not give the solution.

Now, applying second boundary condition

$$\psi = 0,\ \text{At } x=L$$

$$0 = A\sin KL$$

Either A=0 or sin KL=0

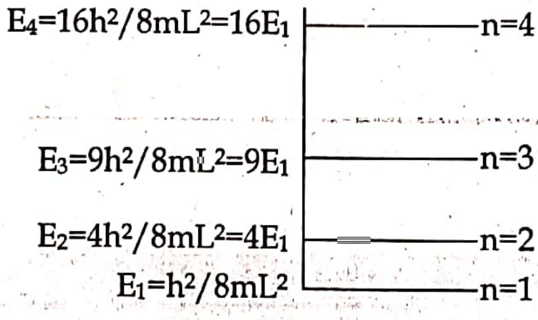
A $\neq 0$, if A = 0 entire function will be zero as B = 0.

$$\therefore \sin KL = 0 \text{ or } KL = n\pi \quad (n = 1, 2, 3, \ldots\ldots\ldots)$$

$$K = \frac{n\pi}{L} \Rightarrow K^2 = \frac{n^2\pi^2}{L^2}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2} \Rightarrow E_n = \frac{n^2\pi^2 h^2}{2mL^2 \times 4\pi^2} = \frac{n^2 h^2}{8mL^2}$$

This relation shows that energy is quantized. ($E_n \alpha\ n^2$)



The Eigen function is

$$\psi(x) = A\sin Kx = A\sin\frac{n\pi}{L}x$$

To find the value of A, we can apply the normalizing condition

$$\int_{-\infty}^{\infty} |\psi(x)|^2\, dx = 1$$

$$\int_0^L A^2 \sin^2\frac{n\pi}{L}x\,dx = 1$$

$$A^2\int_0^L \frac{1}{2}\left(1 - \cos\frac{2n\pi}{L}x\right)dx = 1$$

$$\frac{A^2}{2}\left[x - \frac{L}{2n\pi}\sin\frac{2n\pi x}{L}\right]_0^L = 1$$

$$\frac{A^2 L}{2} = 1 \text{ or } A^2 = \frac{2}{L} \text{ or } A = \sqrt{\frac{2}{L}}$$

$$\boxed{\psi(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi}{L}x}$$

Operators :

Plane monochromatic wave is described by the wave function.

$$\psi(x,t) = A e^{i(kx-\omega t)} \quad \text{--- (1)}$$

We know that $\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda} = \frac{hk}{2\pi} = \hbar k \Rightarrow p = \hbar k$ --- (2)

Also, $E = h\nu = \frac{hc}{\lambda} = \frac{hck}{2\pi} = \hbar ck = \frac{\hbar c 2\pi}{\lambda} = \hbar 2\pi\nu = \hbar\omega$

$$\Rightarrow \quad E = \hbar\omega \quad \text{--- (3)}$$

Using eqn (2) & (3) in eqn (1) becomes -

$$\psi(x,t) = A\exp\left[\frac{i}{\hbar}(px - Et)\right] \quad \text{--- (4)}$$

Diff. eqn (4) w.r.t. t & x we get -

$$\frac{\partial\psi}{\partial t} = A e^{\frac{i}{\hbar}(px-Et)} \cdot \left(-\frac{iE}{\hbar}\right) = -\frac{iE}{\hbar}\psi$$

Now $i\hbar\frac{\partial\psi}{\partial t} = i\cancel{\hbar}\frac{-iE}{\cancel{\hbar}}\psi = E\psi \Rightarrow i\hbar\frac{\partial\psi}{\partial t} = E\psi$ --- (5)

$$\frac{\partial\psi}{\partial x} = A e^{i/\hbar(px-Et)} \cdot \left(\frac{ip}{\hbar}\right) = \frac{ip}{\hbar}\psi \Rightarrow -i\hbar\frac{\partial\psi}{\partial x} = p\psi \quad \text{--- (6)}$$

From eqn (5) & (6) $\boxed{E \rightarrow i\hbar\frac{\partial}{\partial t}}$ and $\boxed{p \rightarrow -i\hbar\frac{\partial}{\partial x}}$

An operator is a rule by means of which a given function is changed into another function.

We know that $H\psi = E\psi$

Time independent form of E_{op} is $E_{op} = H = \frac{\hbar^2}{2m}\nabla^2 + V$

Time dependent Schroedinger eqn is -

$$H\psi = i\hbar\frac{\partial\psi}{\partial t} \Rightarrow (H)\psi = \left(i\hbar\frac{\partial}{\partial t}\right)\psi$$

Time dependent value of Hamiltonian is $\boxed{H = i\hbar\frac{\partial}{\partial t}}$

Thus we have $\boxed{E_{op} = H = -\frac{\hbar^2}{2m}\nabla^2 + V = i\hbar\frac{\partial}{\partial t}}$

Momentum operator: $H = K.E + P.E = \frac{p^2}{2m} + V$.

Also $H = -\frac{\hbar^2}{2m}\nabla^2 + V$

$\therefore \frac{p^2}{2m} + V = -\frac{\hbar^2}{2m}\nabla^2 + V \Rightarrow p^2 = -\hbar^2\nabla^2$

$P_{op}^2 = -\hbar^2\nabla^2 \Rightarrow P_{op}^2 = i^2\hbar^2\nabla^2 \Rightarrow P_{op} = i\hbar\nabla$

$P_{op}^2 = \frac{\hbar^2}{i^2}\nabla^2 \Rightarrow \boxed{P_{op} = \frac{\hbar}{i}\nabla}$

Velocity operator: $P_{op} = m V_{op} = \frac{\hbar}{i}\nabla$

$\boxed{V_{op} = \frac{\hbar}{im}\nabla}$

Kinetic energy operator: $H = KE + PE = T_{op} + V_{op}$

$H = -\frac{\hbar^2}{2m}\nabla^2 + V$.

$\boxed{T_{op} = -\frac{\hbar^2}{2m}\nabla^2}$ $\boxed{V_{op} = V}$

Ques-1. An operator $\frac{d^2}{dx^2}$ has eigen function $\psi = e^{ax}$, Calculate eigen value.

$\hat{O}\psi = o\psi$ where $\hat{O}$ is operator and o is eigen value.

$\hat{O} = \frac{d^2}{dx^2}$ and $\psi = e^{ax}$

$\hat{O}\psi = \frac{d^2}{dx^2}e^{ax} \Rightarrow \hat{O}\psi = \underbrace{a^2}_{\text{eigen value}} e^{ax}$

Relation between group Velocity and Phase Velocity for a non relativistic free particle :-

According to de-Broglie hypothesis - $\lambda = \frac{h}{m v_g}$

Total energy $E = \frac{1}{2} m v_g^2$

Also, $E = h\nu \Rightarrow \nu = \frac{E}{h} = \frac{m v_g^2}{2h}$

$\therefore$ Phase velocity, $v_p = \nu \cdot \lambda = \frac{\cancel{m} v_g^{\cancel{2}}}{2\cancel{h}} \times \frac{\cancel{h}}{\cancel{m} \cancel{v_g}} = \frac{v_g}{2}$

$$\boxed{v_p = \frac{v_g}{2}}$$

Hence for a non-relativistic free particle, the phase velocity is half of the group velocity.